

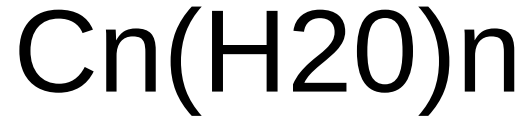
TOPIC;BIO-MOLECULES

Lecture 3

SUB TOPIC 4: CARBOHYDRATES BIOMOLECULE

- **Learning Objectives**

- Define carbohydrates in chemical terms
- Classify carbohydrates in to three major groups with examples of each group
- List the monosaccharides of biological importance and learn their properties
- List the disaccharides of biological importance and learn their properties
- List the polysaccharides of biological importance and learn their properties
- Study the chemistry and functions of glycoproteins



- **Main source of energy** : providing energy= 4kcal/g . Carbohydrates are based upon the general formula $\text{C}_n(\text{H}_2\text{O})_n$
- – As if they were hydrates of carbon (most abundant organic molecules in our biosphere)

Functions of Carbohydrates

- – **Energy source** (glucose)
- – **Energy storage** (glycogen, starch)
- – **Carbon source** (pyruvate used to make Ile, Leu, Val, Ala)
- – **Structure/Protection** (chitin, cellulose, connective tissue)
- – **Recognition/Signaling** (Antibodies used for immune system recognition)
- – Can be **attached** to other **macromolecules** (glycoproteins and glycolipids)

Classes of carbohydrates

- Monosaccharides (simple sugars like glucose)
- Disaccharides (sucrose)
- Trisaccharides
- Polysaccharides (oligosaccharides) (starch, cellulose, glycogen) – long chains of monosaccharides; chains of monosaccharides bridged through oxygen atoms – can be thousands long and can be branched

TYPES

- **A. Simple carbohydrates** 1. Sugars (Simple carb.) : Monosaccharides : The most abundant monosaccharides in nature are the 6-carbon sugars i.e glucose, Fructose;
- • Formula **$C_n(H_2O)_n$** only holds for monosaccharides
- Common monosaccharides contain from 3 to 6 carbon atoms— these sugars are called **trioses, tetroses, pentoses, hexoses**

b. Disaccharides

- When two monosaccharides are covalently bonded together by glycosidic linkages a disaccharide is formed. Glycosidic bond is formed when the hydroxyl group on one of the sugars reacts with the anomeric carbon on the second sugar.
- Biologically important disaccharides are sucrose, maltose, and Lactose.
- Maltose contains two D glucose residues
- Lactose is a disaccharide of β -D galactose and β -D-glucose
- Sucrose is a disaccharide of α - D- glucose and β -D-fructose.

B. Complex Carbohydrates (Polysaccharides)

- There are two types of polysaccharides .These are:
- **Homopolysaccharides** that contain only one type of monosaccharide building blocks. Example of Homopolysaccharides: **Starch, glycogen, Cellulose and dextrans.**
- **Dextrans**-Since they do not easily go out of vascular compartment they are used for intravenous infusion as plasma volume expander in the treatment of hypovolumic shock.

B. Complex Carbohydrates (Polysaccharides)

- **Heteropolysaccharides**, which contain two or more different kinds monosaccharide building blocks. Example; **Glycosaminoglycans, (GAGs or mucopolysaccharides)**
- 1. **Glycosaminoglycans, (GAGs or mucopolysaccharides)**-They are long, usually unbranched, composed of a repeating disaccharide units

Heteropolysaccharides cont....

Function of Glycosaminoglycans. (GAGS)

1. They have the special ability to bind large amounts of water, thereby producing the gel-like matrix that forms the basis of the body's ground substance.
2. Since they are negatively charged, for example, in bone, glycosaminoglycans attract and tightly bind cations like Ca^{++} , they also take-up Na^{+} and K^{+} .
3. GAGs stabilize and support cellular and fibrous components of tissue while helping maintain the water and salt balance of the body.
4. Its essential components of the extra cellular matrix, GAGs' play an important role in mediating cell-cell interactions.

Heteropolysaccharides cont....

- **2. Ground substance** is a part of connective tissue, which is a gel like substance containing water, salt, proteins and polysaccharides.

An example of specialized ground substance is the **synovial fluid**, which serves as a lubricant in joints, and tendon sheaths.

Heteropolysaccharides cont....

3. Heparin:

- Contains a repeating unit of **D-glucuronic** and **D-gluconsamine**, with sulfate groups on some of the hydroxyl and aminx-groups
- It is an important anticoagulant, prevents the clotting of blood by inhibiting the conversion of **pro-thrombin** to **thrombin**.
- Thrombin is an enzyme that acts on the conversion of plasma **fibrinogen** into the **fibrin**.
- It is found in mast cells in **lung, liver skin and intestinal mucosa**.

Heteropolysaccharides cont....

4. Glycoproteins (Mucoproteins)

- Glycoproteins are proteins to which oligosaccharides are covalently attached.
- Oligosaccharides can also be attached to proteins. Through glycosidic linkages to **serine, threonine or asparagines**

Functions:

- a. Influence structure, folding and stability of protein
- b. May determine the lifetime of a protein (mark protein for age)
- c. Serve as markers to identify a cell type

Heteropolysaccharides cont....

d. When glycosylated proteins are at the cell surface:

Can **modulate cell-cell interactions**

e. Changes in carbohydrate content may influence contact inhibition of cells

Can **modulate cell – molecule interactions** (e.g. hormone w/receptor)

Can serve as **antigenic determinants** (how antibody recognizes the protein) on proteins

Heteropolysaccharides cont....

Proteoglycans

- When glycosaminoglycans are attached to a protein molecule the compound is called proteoglycan [proteoglycans = Glycosaminoglycans + proteins]
- **BLOOD TYPES AND GLYCOSYLATION**
- Presence or absence of the terminal carbohydrate is genetically determined and determines the blood type.
- Blood plasma contains antibodies against foreign blood-group antigens that aggregate the foreign blood cells

Heteropolysaccharides cont....

- Type A blood has antibodies that recognize B sugars
- Type B blood contains antibodies against A sugars
- Type O blood has antibodies against both A and B sugars (universal donor)
- Type AB blood contains neither antibody (universal acceptor)
- Incompatible blood types cause precipitation of RBCs, block blood flow in organs and can cause death
- Also influenced by the presence or absence of Rh factor (blood protein) cell proteins

SUB TOPIC 5: WATER MOLECULE

- **Learning Objectives**
 - Define **Water** in chemical terms
 - Explain the biological importance of **Water** and learn their properties
 - Explain **the role of Water** in **biochemical processes**

Introduction

Regulates many processes in the body.

- **Homeostasis** ◦ Process of maintaining a steady state in the body.
- **Dehydration** ◦ Severe reduction in the body's water content. ◦ Condition that results in the body's loss of important electrolytes.

Water

Water is the solvent of choice for biological systems

- Constitutes 70-85% of cell weight, typically
- Important as a **solvent** and a **reactant** in biochemical reactions
- Helps regulate **temperature** since it is able to absorb large amounts of heat
- Helps regulate **intracellular pH**
- Used for transport – delivers nutrients and removes waste from cells

Water

- is a unique solvent whose properties are extremely important to biochemistry.
- - In water, the hydrogen atoms have a partial positive charge,
- - The polar nature and geometry of the water molecule allows water molecules to form hydrogen bonds with each other and with dissolved hydrophilic substances.
- - Hydrogen bonds between water molecules= electrostatic attraction between the oxygen atom of one water and the hydrogen of another

SUB TOPIC 5: VITAMIN MICROMOLECULE

- **Learning Objectives**

- Define vitamins in chemical terms
- Classify Vitamins into two major groups with examples of each group
- List the Fat soluble Vitamins of biological importance and learn their properties
- List the **Water soluble** Vitamins of biological importance and learn their properties
- Explain biological importance/functions of each Vitamin.

Introduction

- • Human body cannot synthesize it except for vitamin D and K.
- • Important for proper body functioning.
- • Unlike carbohydrates, fats, and proteins, vitamins DO NOT provide energy (calories).

TYPES OF VITAMINS:

- **FAT SOLUBLE**- A,D,E,K • Fatsoluble vitamins can be stored in the body while water- soluble vitamins are not and get easily excreted in urine
- **WATER SOLUBLE**- B and C • Vitamins B-complex and C are heat labile vitamins and are easily destroyed by heat, air or during drying, cooking and food processing.

Vitamin A

- Ample reserves of fat soluble vitamins are stored in the tissues as they are not readily absorbed from the food. With the exception of Vit. K, they do not serve as coenzymes. Indeed Vit D act more like hormone.
- **Vit A deficiency leads to ;**
- **Carrier protein.** Low plasma [Vitamin A] has been shown to be associated with an increased risk of developing cancer.
- Failure of bone formation (Thick, solid bones).

Vitamin A

- **Effect on Skin;** The deficiency causes dryness and roughness of skin developing keratosis of hair follicles with concomitant deficiency of Vit-B complex.
- **Effect on Bone and teeth** Bone growth is markedly impaired. Osteoclastic activity is also hampered, causing defective bone formation
- **Effect on general Metabolism** Zinc is necessary to maintain normal plasma concentration of Vit A. This vitamin is also necessary for the conversion of trioses to glucose perhaps indirectly through adrenal cortex that synthesizes hormones concerned with Gluconeogenesis.

Vitamin -D

- **The sterol, 7-dehydrocholesterol**) Vitamin D is the only vitamin that is usually not required in the diet, for this reason it is rather classified as a hormone since under conditions of inadequate exposure to sunlight that dietary intake is required.
- Calciferol, cholecalciferol (D3) Bound to specific D-binding protein, chloecalciferol moves via circulation directly to the liver.
- The 25 (OH)-chloecalciferol is a potent Vit.D3 and is also produced in a smaller proportion in the kidney.

Vitamin -D

- Vit D3 is also found in the diet where its absorption is associated with other fats, and is transported to the liver by chylomicrons.
- A significant proportion of 25 (OH)-D3 is excreted in the bile and is reabsorbed in the small bowel, producing an enterohepatic circulation.
- Hypocalcemia, Phosphatemia stimulates the release of PTH which enhances the synthesis of Vitamin D. On the other hand, high calcium, phosphate inhibit the synthesis.

Vitamin -D

Functions:

- Target organs are bone, Kidney and Intestine. Calcitriol promotes bone mineralization.
- **Intestine:** This vitamin promotes absorption of calcium, phosphates. The mechanism of action resembles that of steroid hormone. It crosses cell membrane, binds to cytoplasmic receptor to form a complex, which is translocated to the nucleus. Here it binds to chromatin, induces the synthesis of calcium binding protein. Thus calcium absorption is stimulated. Defect in cytoplasmic receptor may lead to Rickets.

Vitamin -D

- **Kidney:** Reabsorption of calcium, phosphate are enhanced.
- **Bone:** It promotes synthesis of osteocalcin which is needed for bone mineralization. It also
- promotes bone collagen synthesis.

Catabolism of Vit D:

- Hydroxylation at C26 takes place in Liver by oxidases .This compound has no Vitamin D
- activity.

VITAMIN K

There are three types, Menaquinone (K2)present in animals ,Phylloquinone (K1) present in Plants. Menadione a synthetic water soluble vitamin is available for treatment.;**K K1 & K2**

Functions

- It is the only one acting as co-enzyme from the group of Fat soluble vitamins.
- This vitamin is also synthesized by intestinal bacteria. It is required for post translational modifications of several proteins required in the coagulation cascade. For e.g. Factor II, VII. IX and X .
- Activation is carried out by the carboxylation of specific glutamate residues on the prothrombin by Vit K dependent enzyme.

(B) Water Soluble Vitamins

- **The B- Vitamins**
- Since they are water soluble excess can be excreted through urine.
- Hyper-vitaminosis may not cause toxicity. Most of these vitamins act as coenzymes.
- The B- Vitamins are essential and must be provided through diet: these include:

(B) Water Soluble Vitamins

- - Thiamine (Vit B1)
- - Riboflavin (Vit B2)
- - Niacin (Nicotinic acid (or Nicotinamide)
- - Pantothenic acid (Vit B5)
- - Vitamin B6 (Pyrodoxine,pyridoxal,& Pyridoxamine)
- - Biotin
- - Vitamin B12 (Cobalamin)
- - Folic Acid

VITAMIN- B2

- **Riboflavin**, also known as **vitamin B2**, is a component of the flavin coenzymes, **FAD** and **FMN**.
- It is composed of an isoalloxazine ring system linked to ribitol.
- The ability of the ring system of **riboflavin** to exist as a semiquinone allows the flavin coenzymes to accept electrons either singly or in pairs. **NAD⁺** and **NADP⁺** can **only** accept electrons in pairs.
- It is mainly used in the energy metabolism of Sugars and Lipids. The activation of FMN and
- FAD is an ATP-dependent.

VITAMIN- B3

- Niacin is not a vitamin in a strictest sense of the word, since it can be synthesized from **Tryptophan**.
- However, conversion of **Tryptophan** to **Niacin** is relatively inefficient (60 mg of
- Tryptophan is required to produce 1mg of Niacin) and occurs only after all the body requirements for Tryptophan is met. Thus most people require dietary sources of both **Tryptophan** and **Niacin**.
- Niacin contains a substituted Pyridine ring and when gets activated forms **NAD⁺** and its phosphorylated derivative is **NADP⁺** .which are co enzymes of many dehydrogenases.

VITAMIN- B 6 (Pyridoxine)

- **Exists in three forms:** Pyridoxine, Pyridoxal & pyridoxamine and their corresponding phosphates.
- **Pyridoxal phosphate (PLP)-** Pyridoxal phosphate participates in transaminations, decarboxylations, racemizations, and numerous modifications of amino acid side chains.
- It is also used for the synthesis of Neurotransmitter, Serotonin and Nor-Adrenalin.
- Used as a component of Sphingolipids necessary for myelin formation and Heme synthesis as well.
- Hypochromic microcytic anemia since PLP is required for Heme synthesis.
- Deficiency in infants cause convulsions due to inactive glutamate decarboxylase, GABA not formed there by impaired neurotransmission.

VITAMIN-B5

- Pantothenic **Acid** (Vit B 5) Coenzyme A.
- **Pantothenic acid** is a vitamin that forms an essential part of the acyl-carrier moiety, **coenzyme A**.
- **Coenzyme A** (A for acyl) participates in the activation of acyl groups in general, including the acetyl group derived from **pyruvate**.
- The coenzyme is derived metabolically from **ATP**, the vitamin **pantothenic acid**, and **-mercaptoethylamine**.

VITAMIN-C

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The hydroxylation reactions in collagen involve **vitamin C**
In general hydroxylation reactions require Vit C.

Example: Hydroxylation of cholesterol.

Functions:

- • Collagen biosynthesis
- • Degradation of Tyrosine
- • Absorption of Iron
- • Steroidogenesis
- • Adrenaline synthesis
- • Bile acid formation
- • Degradation of tyrosine
- • Bone mineral metabolism
- • Potent anti oxidant